

AI Exposure at Three Levels: Reconciling Occupation, Task and Skill Measures of the Labour Market

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Extended summary

This paper proposes a way of putting the leading measures of exposure to generative artificial intelligence onto a common footing, and reports what becomes visible once they are. The motivation is a mismatch that the applied literature has largely passed over. Exposure is measured at three different levels of the same labour market, and the measures are then compared, combined and fed into models as though they described the same object. Felten, Raj and Seamans (2021) score the fifty-two abilities of the Occupational Information Network. Eloundou, Manning, Mishkin and Rock (2023) score its task statements. Srinivasan, Chen and Zakerinia (2024) score occupations. Each is used in applied work at whatever level the application happens to require, and the aggregation or disaggregation that carries it there is rarely examined and never validated.

The premise underlying all three is common ground and is not claimed here as novel. A firm produces output, that output is produced by activities being carried out, and those activities are carried out by combinations of human capability, so that occupations, tasks and skills are three descriptions of a single process and a technology entering production must be traceable through all of them. Autor, Levy and Murnane (2003) and Acemoglu and Autor (2011) established the task layer, the skill layer descends from Katz and Murphy (1992), and Freund and Mann (2026) have developed the general equilibrium version in which occupations bundle tasks, workers hold task-specific skills, and automating a task raises the weight on those that remain. Their model, estimated on individual worker data, projects that the return to social and non-routine manual skills rises while the return to analytical skills falls.

What that literature needs, and does not have, is an account of the exposure measures it consumes. Freund and Mann require a mapping from technology to automated tasks and take it from one source. There are several such sources, they were built at different levels by different teams, and they do not agree with one another. The contribution of this paper is to establish what each of them measures, whether any of them survives being moved between levels, and what the underlying data will bear, before any of them is fed into a model of the three layers.

Four things follow. The propagation that carries a task-level measure down to skills can be validated from outside, because two of the measures reach the same quantity by genuinely

independent routes, and it reaches the reliability ceiling of the source it carries rather than adding error of its own. The capability measures are considerably less independent of one another than their mutual agreement suggests, since the Srinivasan automation score applies the Eloundou rubric to the same task descriptions, so their correlation of 0.833 across occupations demonstrates that the procedure reproduces across data vintages and model versions rather than that separate approaches converge. Augmentation, as currently measured, is largely a transformation of automation rather than a second measurement, being defined as a Herfindahl index over the same exposure classification and therefore explained to 77 percent by a quadratic in the automation score; what remains of it does carry skill structure, with social skills, arts and humanities, work output, engineering and psychomotor abilities loading on the augmenting side, but the automating side of that contrast is mostly the algebra. And the natural design for asking whether any of this has reached realised wages cannot answer the question, because every deviation from the pre-2020 path in these series begins in 2020 or 2021, before large language models arrived, so that the pandemic and the technology cannot be told apart in United States occupational data.

The instrument used throughout is not well suited to the purpose, and the paper is explicit about that rather than apologetic. Its importance ratings compress towards uniformity whenever they are converted into shares, with consequences documented in Section 4 that reach as far as published valuations of tasks. Its descriptors are collinear enough that the effective dimension of the space is roughly a third of its nominal dimension. Its occupational content is revised too slowly to register a job changing while keeping its name, which is very likely how this technology operates. The instrument is used anyway, here and by everyone else, because it is free, it covers the whole economy, and it is the common vocabulary in which all of the measures discussed below are expressed, whereas the alternatives that would serve better are restricted individual data that most researchers cannot obtain. A public instrument whose limits are established is more useful to a field than a private one that few can run, and establishing those limits is therefore part of the work rather than a caveat attached to it.

Introduction

The arrival of generative artificial intelligence has produced a rapid empirical literature applying the task framework, and a smaller set of measurement papers on which that literature depends. The measures have been compared with each other, and their disagreements noted, most often as a puzzle about whether capability-based scores or observed-usage scores are the more informative. What has attracted less attention is that they were not collected at the same level of description, so that comparing them at all requires an act of translation whose properties nobody has examined.

The translation matters because the levels differ in what they can support. Wages are observed once per occupation, and there are eight hundred and twenty-nine of them in the 2025 Occupational Employment and Wage Statistics, carrying a wage bill of 10.70 trillion dollars. Every other quantity in this literature is inferred from those observations. There are 18,796 task

statements, so any set of task values has more unknowns than facts, will always fit whatever it is fitted to, and cannot be contradicted by the data that produced it. There are one hundred and sixty-one descriptors, and after grouping there are twenty-five, so a set of skill values has fewer unknowns than facts and can fail. Whether a claim in this field is testable is therefore settled by how many task statements the Network’s analysts happened to write, which is a property of a survey instrument rather than of an economy, and this is worth stating plainly because it is not a technicality. Freund and Mann cluster roughly nineteen thousand tasks into thirty-eight categories before estimating anything, and that clustering is what makes their estimation possible rather than a convenience of exposition.

This paper is organised around that observation. Section 1 sets out the three layers and what each is for. Section 2 places each exposure measure at the level where it was collected and explains what reaching the other levels requires of it. Section 3 reconciles them, and reports the validation of the step that carries tasks down to skills. Section 4 documents what the underlying data will and will not support. Section 5 applies the apparatus to realised wages and employment across fourteen years, and reports both a null and the reason the null cannot be interpreted as the literature would like.

Section 1: Three descriptions of one labour market

The three layers are not interchangeable views of one thing, and treating them as though they were is what produces most of the difficulty in this literature. They are better understood as performing three different functions, none of which the others can perform.

The wage bill is a budget. Everything must sum to it, which disciplines a description without testing it, because quantities constructed as shares of a total will return that total whatever values the shares take. This is worth dwelling on, because it is the reason a reconciliation to the national wage bill, which several exercises in this area report as a validation, establishes only that the arithmetic ran.

Tasks are where the technology has been observed. Every measurement of generative artificial intelligence, without exception, was made by examining tasks or the activities that compose them, because that is the level at which a capability can be judged against a description of work. No account of this technology can dispense with the task layer, whatever the identification difficulties attached to it.

Skills are where a claim can fail. Being fewer than the occupations whose wages are observed, they are the only level at which a set of values is over-determined by the data, and therefore the only level at which the framework can be wrong in a way the data will reveal. The grouping used here is the third level of the Network’s own content model, which supplies twenty-five named groups spanning abilities, skills, knowledge and work activities, and which has the advantage of removing the choice of grouping from the analyst’s discretion.

Figure 1 sets out the three layers and the measures that enter at each.

Section 2: Measures collected at three levels

Felten, Raj and Seamans (2021) constructed their exposure measure by mapping artificial intelligence applications onto the fifty-two abilities of the Network and then aggregating to occupations, and they published the intermediate ability-level scores as well as the occupational ones. Because those abilities are fifty-two of the one hundred and sixty-one descriptors used here, the measure enters the skill layer directly, requiring no propagation and offering nothing that needs validating. All fifty-two join once a single renaming between releases is handled. Applied work almost always uses the occupational aggregate and discards the ability-level file, which is a loss, since it is the only exposure measure that arrives natively at the level where claims can be tested.

Eloundou, Manning, Mishkin and Rock (2023) rate 19,265 task statements for exposure, both by human raters and by a large language model, of which 18,179 join the task set used here. This is the measure Freund and Mann employ, and it is the richest of the four, being complete in its coverage and carrying two independent ratings of the same objects. Reaching skills from tasks requires the Network’s crosswalk from task statements to detailed work activities, which roll up to the forty-one generalized activities and thence, through the published crosswalks from skills and abilities to work activities, to the wider descriptor set. That step is where this framework is most exposed, and Section 3 tests it.

Srinivasan, Chen and Zakerinia (2024) score automation and augmentation separately for nine hundred and eleven occupations, of which nine hundred and ten join the Network by name, yielding seven hundred and eighty-five six-digit occupations and covering six hundred and eighty-one of the six hundred and ninety-three in the panel used here. Theirs is the only source offering a separate augmentation figure across the whole economy, and Section 3 examines how separate it actually is. Both of their indices are built from task descriptions and a language model’s classification of them, with job postings entering only to validate the indices and never to construct them, so neither score contains labour market outcomes and both may be used to explain outcomes without circularity. Reaching skills from occupations requires an inversion rather than an aggregation, since the score is a combination of skill exposures and must be decomposed rather than averaged.

The Anthropic Economic Index sits at the task layer, observes usage rather than capability, and separates automation from augmentation, which would make it the most valuable of the four were it usable as an exposure measure. It is not. It records only tasks for which the model has actually been used, covering 20.9 percent of the tasks considered here, and the covered tasks are more exposed by 0.236 on a zero to one scale than those it omits, with the most used tenth accounting for three quarters of all recorded usage. Placing it on the right hand side of any regression would be selecting on the quantity to be explained. It remains informative as an outcome within the subset it covers, and that is how Section 5 treats it.

Section 3: Reconciling the measures

The step that carries a task-level measure down to skills is assumed in every applied use of these data and has not previously been tested, for the reasonable reason that testing it requires an independent measurement of the same quantity at the destination level. That measurement exists. Felten and co-authors measure ability exposure directly, and propagating the Eloundou ratings through the crosswalk reaches ability exposure by a route that shares no data with theirs. Across the fifty-two abilities the two correlate 0.533 by Pearson and 0.632 by Spearman.

The benchmark against which those figures should be read is not unity. Eloundou’s own human and machine raters agree on 65.6 percent of task labels, and their numeric scores correlate 0.591, so the propagation recovers as much of the underlying signal as the source contains and adds little error in transit. Variation also survives the aggregation, which was not assured, since averaging eighteen thousand task ratings into twenty-five groups could easily have arrived at something uniform. It does not: the coefficient of variation is 0.292 across a range from 0.045 to 0.513, and the resulting order places mental processes at the top, foundational literacy and numeracy close behind, and physical abilities at the bottom, which is what an account of language models would predict without any tuning.

Turning to the measures themselves, the automation score of Srinivasan and co-authors correlates 0.833 with the Eloundou exposure aggregated to occupations, and the reason is more instructive than convergent validity would be. Their automation score is the Eloundou construction, applied by the authors to the task descriptions of a different release of the Network using a later version of the same model, with the same rubric distinguishing tasks a language model can substantially accelerate from those it cannot and the same weighting of one and one half applied to the two exposed categories. The correlation therefore establishes that the procedure reproduces across data vintages and model versions, which is worth knowing and is not what agreement between measures is usually taken to show. The literature’s observation that capability-based scores agree with one another needs reading in this light, since several of them share a rubric, a source of task descriptions, or both.

Augmentation requires the same care and has been overstated here in an earlier draft. It is not an independent measurement of complementarity. Srinivasan and co-authors define it as one minus a Herfindahl index over the shares of exposed and unexposed tasks within an occupation, on the reasoning that a job combining automatable work with work that resists automation stands to gain most, which is an assumption about complementarity imposed on the exposure classification rather than a second observation of it. Because that index is a quadratic in the exposed share, and the automation score is a closely related summary of the same classification, a quadratic in automation accounts for 77 percent of the variance in augmentation across occupations. Whatever independent content the measure holds lies in the remaining 23 percent, and any statement contrasting the two dimensions has to be made net of the mechanical part rather than on the raw scores.

Inverting the two occupational scores onto the twenty-five skill groups is better conditioned than the equivalent exercise on wages, because exposure is not a share of the wage bill and therefore

requires no employment weighting, leaving the effective sample at the number of occupations. Skill groups account for occupational exposure with cross-validated fit of 0.712 on automation and 0.415 on augmentation, and the named groups outperform ten principal components on both, so the detail earns its place here in a way that Section 4 shows it does not against wages.

The contrast between the two dimensions has to be taken net of the mechanical relation between them, and doing so changes which part of it stands. On the raw scores, ten of the twenty-five groups show an augmentation loading distinguishable from their automation loading, a separation stable across five hundred bootstrap resamples. Removing the quadratic in automation first and inverting only what is left, eleven groups remain distinguishable, but they are not the same eleven and the two profiles correlate 0.604 rather than coinciding. Social skills, engineering and technology, arts and humanities, work output and psychomotor abilities hold up, with skill groups accounting for 0.209 of the residual variation. Business and management and mental processes do not, falling from statistics of around three to 0.4 and 1.2 respectively, which places them among the artefacts of the formula rather than among the findings.

What can be said, then, is narrower than the raw contrast suggests. Work involving people, physical execution, and applied technical judgement is associated with the complementary side of these measures beyond what the construction mechanically implies, while the apparent concentration of automation on cognitive and administrative content is mostly the algebra of a Herfindahl index restating an exposure share. Figure 2 shows both profiles. The surviving pattern points in the same direction as the Freund and Mann projection and is a good deal weaker than it, describing the residual content of exposure scores rather than estimated returns.

Section 4: What the instrument will bear

Importance in the Network is recorded on a five-point scale of which respondents use a narrow part, so that converting a rating into a share yields something close to uniform, and this occurs wherever the conversion is made. In the median occupation the most important task rates 1.41 times the least important. Where the descriptor mix of a task is set to the average of its occupation, as it is in the standard construction, all 124,614 occupation by descriptor pairs containing more than one task carry a single distinct value, so that the skill layer becomes a fixed linear image of one number per occupation and can redistribute information without adding any. A descriptor proximity matrix built from occupational similarity places 98.5 percent of its 798,342 pairs above 0.90 with a coefficient of variation of 0.022. And an occupation's profile across the forty-one generalized work activities sits a total variation distance of 0.10 from a uniform profile, which means that one of the Network's two descriptions of what a job involves carries almost no information at all, and that the two descriptions accordingly agree at a median rank correlation of only 0.33.

The consequence reaches published quantities and is not confined to parameters. Valuing 17,383 tasks by allocating each occupation's wage bill in proportion to task importance produces values correlating 0.997 on ranks with dividing that same wage bill by the occupation's number of tasks, with eighty-seven of the largest hundred tasks identical under the two procedures, so that a

valuation of this kind restates occupational employment and pay with a task count as its divisor. The Network also publishes the frequency with which each task is performed, distributed over seven bands, which varies about fifteen times more than importance with a median ratio of 14.78, and substituting it lowers the correlation to 0.883 and separates tasks that were previously indistinguishable. Frequency does not make the valuation identified, for the counting reason given in the introduction, but it makes it informative.

What can be estimated divides cleanly. Fitting occupational wages with descriptor intensities and evaluating on occupations held out of estimation gives 0.554 for a single ordinal measure of the preparation a job requires, 0.700 for twenty principal components, 0.648 for the twenty-five named groups, and 0.544 for all one hundred and sixty-one descriptors, so that the full descriptor set predicts held-out occupations worse than one variable while reaching 0.878 within sample. The cause is degrees of freedom rather than specification, since employment weighting, which any decomposition of the wage bill requires, gives an effective sample of one hundred and twenty-one against eight hundred and twenty occupations in the sense of Kish (1965), the twenty largest occupations carrying a third of employment between them. Quantities of a skill group are measurable, being employment multiplied by intensity with nothing fitted. Price levels are not recoverable, since eleven of the twenty-five groups take a negative price in some year and constraining prices to be non-negative pins twenty of the twenty-five at exactly zero, which establishes that the unconstrained estimates were relying on offsetting signs rather than that the constraint was badly chosen. Price changes are recoverable, and this is fortunate, because the effect of a technology is a change rather than a level.

Inference in this setting requires more care than it usually receives. Comparing any two periods in these data returns between five and nine of the twenty-five groups apparently changing course when nothing whatever has happened, so that a result reporting eight is reporting nothing. Conventional standard errors are computed against how precisely a window's average is determined rather than against how much the series moves between ordinary years, and the second quantity is the larger. Composition series move slowly and smoothly enough that a shift worth 0.17 percent of a level per year produces a statistic of 8.2, so magnitude must be reported alongside significance. The procedure adopted throughout is to estimate the quantity of interest for every year transition available, which measures the year-to-year variability directly, and to construct the reference distribution by relabelling which three transitions are designated as treated. Four results in the course of this work reached conventional significance and did not survive that procedure.

Section 5: Evidence on realised outcomes

The apparatus is applied to a balanced panel of six hundred and ninety-three occupations spanning 2012 to 2025, with a wage bill rising from 5.15 to 8.95 trillion dollars, assembled from the Occupational Employment and Wage Statistics and mapped across the 2018 revision of the occupational classification using the crosswalk the Network distributes, which raises coverage of the 2018 wage bill from 0.813 to 0.936 and without which the computing occupations disappear

from the earlier years.

Across 2022 to 2025 no skill group moved by more than 1.51 times its own year-to-year volatility, and five of the twenty-five exceed two of their own standard errors against a median of four across two hundred and eighty-six relabellings, which places the observed period inside the range that ordinary years produce. Quantities agree with prices, at a probability of 0.455 weighted by headcount, and the agreement is worth noting because quantities involve no estimation at all.

A directional test against the Freund and Mann projection, with group membership assigned from their own vocabulary before any outcome was examined, is not supported on wages and moves in the opposite direction, the returns to both social and analytical content having risen against their pre-period paths with analytical rising further. The social against analytical contrast in employment moves as projected, at a probability of 0.021 against all relabellings and 0.273 against consecutive triples, and it is the one specification with no pre-existing trend to confound it.

The more consequential finding concerns what these tests can establish at all. Deviations from the pre-2020 path are shown by year in Figure 3, and every one of them begins in 2020 or 2021. On wages the residual runs at minus 0.005 in 2020 and worsens monotonically to minus 0.025 by 2025, and on employment the series takes an excursion of minus 0.040 in 2020 and remains positive from 2021 onward. Large language models became publicly available in late 2022, after both breaks had begun. The pandemic sits between the pre-period and the treatment period and moves these series by more than the effect being sought, so that a before-and-after comparison in United States occupational data cannot separate the technology from the recovery. This is a property of the design rather than of its implementation here, and it applies to any study drawing the same comparison from the same source.

Two directions follow from it. The pandemic moved which occupations grew, whereas a technology of this kind should move what happens inside an occupation, and the second is a different object that a comparison across occupations cannot see. The validated propagation of Section 3 is what a within-occupation design would require, and the Anthropic index supplies task-level outcomes on the fifth of tasks it covers, where the selection is on tasks rather than on time. Separately, the separation of augmentation from automation reported in Section 3 suggests a shock structure that the current generation of models does not represent, since automating a task and assisting it have different implications for who gains, and the loadings reported there are what a model incorporating the distinction would need.

Conclusion

Three literatures measure exposure to generative artificial intelligence at three different levels of one labour market, and the levels are not interchangeable, because the wage bill constrains without testing, tasks are where the technology has been observed, and skills are the only level at which a claim can fail. This paper builds the apparatus for stating the measures together using only public data, validates from outside the step that carries a task measure down to skills, and finds that the capability measures are close to a single construct while augmentation is a

second dimension they do not carry and which separates from automation across the skill groups in a pattern that is stable and interpretable. It documents four specific respects in which the instrument the field runs on constrains what may be concluded from it, and it reports that the natural design for detecting an effect in realised wages cannot work, because the pandemic break precedes the technological one in every series examined.

The premise, that a technology entering production must be traceable through occupations, tasks and skills together, is not settled by any of this, and it is not the property of this paper to settle. What the paper offers is the apparatus by which the attempt can be made on data that anyone can obtain, and a reasonably precise account of what that data will and will not support.

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